

C Programming

Comprehensive Question Bank — Lectures 1–6 & Revision Materials

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Part	Type	Marks
I	Multiple Choice Questions (MCQ)	50
II	True or False	20
III	Correct the Code	5
IV	Predict the Output	5
Total		80

Part I — Multiple Choice Questions (50 Marks)

Choose the correct answer for each of the following:

Q1. Which header file is necessary for using `printf()` and `scanf()`?

- A.
- B.
- C.
- D.

Q2. What is the execution starting point of any C program?

- A. The first line of code
- B. The `main()` function
- C. Preprocessor directives
- D. The `return 0;` statement

Q3. Which of the following is a valid variable name in C?

- A. `2nd_value`
- B. `total cost`
- C. `_myVariable`
- D. `float$`

Q4. What is the size of a `double` data type in C?

- A. 2 bytes
- B. 4 bytes
- C. 8 bytes
- D. 1 byte

Q5. Which format specifier is used to read or print a single character?

- A. %d
- B. %f
- C. %s
- D. %C

Q6. What does the `const` keyword do when declaring a variable?

- A. Allows the variable to be modified anywhere.
- B. Makes the variable read-only.
- C. Makes the variable an integer automatically.
- D. Deletes the variable after use.

Q7. Which operator is used for the modulus (remainder) operation?

- A. /
- B. \
- C. %
- D. &

Q8. If `int x = 10;`, what is the value of x after `x += 5;`?

- A. 10
- B. 5
- C. 15
- D. 50

Q9. Which logical operator represents "Logical OR"?

- A. &&
- B. !
- C. ||
- D. &

Q10. In C, what value is considered "False" in logical conditions?

- A. 1
- B. Any non-zero value
- C. 0
- D. -1

Q11. Which statement executes code only if a condition is true?

- A. `while`
- B. `if`
- C. `switch`
- D. `return`

Q12. The `else` statement is used to:

- A. Create a new loop.
- B. Execute code when the `if` condition is false.
- C. Exit the program.
- D. Define a constant.

Q13. What is the purpose of the `break` statement in a switch case?

- A. To start the next case.
- B. To jump out of the switch block.
- C. To repeat the current case.
- D. To terminate the entire program.

Q14. Which loop is guaranteed to run at least once?

- A. `for`
- B. `while`
- C. `do-while`
- D. None of the above

Q15. What is the result of `5 / 2` when using integers in C?

- A. 2.5
- B. 2
- C. 3
- D. 0

Q16. Which escape sequence is used for a new line?

- A. `\t`
- B. `\b`
- C. `\n`
- D. `\\`

Q17. To get a string that contains spaces from the user, which function is best?

- A. `scanf()`
- B. `printf()`
- C. `fgets()`
- D. `getchar()`

Q18. What is the correct syntax for an infinite `while` loop?

- A. `while(0)`
- B. `while(1)`
- C. `while()`
- D. `while(false)`

Q19. Which operator is used to get the memory address of a variable?

- A. `*`
- B. `&`
- C. `%`
- D. `^`

Q20. How is a string terminated in C?

- A. With a period `.`
- B. With a semicolon `;`
- C. With a null character `\0`
- D. With a new line `\n`

Q21. What happens when a `continue` statement is executed in a loop?

- A. The loop terminates immediately.
- B. The current iteration is skipped.
- C. The program exits.
- D. The loop restarts from the beginning.

Q22. Which data type is most suitable for storing a GPA like 3.75?

- A. `int`
- B. `char`
- C. `float`
- D. `long`

Q23. Which of the following compares if `a` is equal to `b`?

- A. `a = b`
- B. `a == b`
- C. `a === b`
- D. `a != b`

Q24. What is the output of `printf("%d", 10 > 5);`?

- A. True
- B. 10
- C. 1
- D. 0

Q25. A `switch` expression must result in which type?

- A. `float` or `double`
- B. `int` or `char`
- C. string only
- D. Any data type

Q26. Which part of the `for` loop is executed after the body?

- A. Initialization
- B. Condition
- C. Increment/Decrement
- D. None

Q27. What is the ASCII value of the character `'A'`?

- A. 97
- B. 48
- C. 65
- D. 1

Q28. Multi-line comments in C start with:

- A. `//`
- B. `/*`
- C. `#`
- D. `--`

Q29. If `int a = 5;`, what is the value of b in `int b = ++a;`?

- A. 5
- B. 6
- C. 4
- D. 0

Q30. Which operator has the highest precedence?

- A. +
- B. *
- C. ()
- D. =

Q31. `scanf("%d", #);` — What does the `&` represent?

- A. Logical AND
- B. Pointer
- C. Address of the variable
- D. Value of the variable

Q32. The ternary operator `?:` acts as a replacement for:

- A. A `for` loop
- B. A simple `if-else`
- C. A function call
- D. `switch-case`

Q33. What is the size of a `char` variable in C?

- A. 1 byte
- B. 2 bytes
- C. 4 bytes
- D. 8 bytes

Q34. Which keyword is used to return a value from a function?

- A. `give`
- B. `back`
- C. `return`
- D. `exit`

Q35. What is a "Nested Loop"?

- A. A loop that never ends.
- B. A loop placed inside another loop.
- C. A loop that uses `if` statements.
- D. A loop that runs only once.

Q36. To print a `double` with 2 decimal places, you use:

- A. `%d`
- B. `%.2f`
- C. `%lf`
- D. `%2f`

Q37. Which of the following is NOT an arithmetic operator?

- A. %
- B. *
- C. &&
- D. -

Q38. In a `switch` statement, what happens if no case matches and no `default` is provided?

- A. The program crashes.
- B. The program skips the switch entirely.
- C. It repeats the first case.
- D. It generates a compiler error.

Q39. What is the result of `!(5 > 2)`?

- A. 1
- B. 0
- C. True
- D. 5

Q40. Which function is used to find the size of a data type in bytes?

- A. `len()`
- B. `count()`
- C. `sizeof()`
- D. `malloc()`

Q41. What is the maximum number of `else if` blocks allowed after an `if`?

- A. 1
- B. 10
- C. 255
- D. No fixed limit

Q42. If `int x = 5;`, what is the output of `printf("%d", x--);`?

- A. 4
- B. 5
- C. 6
- D. Error

Q43. Which of these is a preprocessor directive?

- A. `int x;`
- B. `main()`
- C. `#include`
- D. `return 0;`

Q44. A `do-while` loop ends with:

- A. A colon :
- B. A semicolon ;
- C. A bracket }
- D. Nothing

Q45. Which header file is used for mathematical functions like `sqrt()` or `pow()`?

- A.
- B.
- C.
- D.

Q46. What is the correct way to declare a constant for PI?

- A. `const float PI = 3.14;`
- B. `float const PI 3.14;`
- C. `constant PI = 3.14;`
- D. `PI = 3.14;`

Q47. In the `for` loop `for(i=0; i<10; i++)`, how many times does it run?

- A. 9
- B. 10
- C. 11
- D. Infinite

Q48. Which format specifier is used for a string?

- A. `%c`
- B. `%d`
- C. `%s`
- D. `%f`

Q49. What does `int main(void)` indicate?

- A. The function returns nothing.
- B. The function takes no arguments.
- C. The program is invalid.
- D. The program will run forever.

Q50. Who are the instructors for this C programming course?

- A. Dr. Dina & Dr. Ghada
- B. Dr. Ahmed & Dr. Sara
- C. Prof. John & Dr. Mary
- D. None of the above

Part II — True or False (20 Marks)

Write (T) for True and (F) for False:

1. C is a case-sensitive language.
2. The `main()` function is optional in a C program.
3. Comments are ignored by the C compiler.
4. A variable name can start with a number (e.g., `1variable`).
5. The `=` operator is used to compare two values.
6. Every statement in C must end with a semicolon `;`.
7. A `while` loop checks the condition after executing the body.
8. `float` and `double` are used to store decimal numbers.
9. The `default` case in a `switch` statement is mandatory.
10. `\t` is the escape sequence for a horizontal tab.
11. `int x = 5.9;` will store the value 5 in x.
12. Logical `&&` returns true if both conditions are true.
13. `scanf()` requires the address operator `&` for string variables.
14. The `break` statement can be used inside an `if` statement (outside a loop).
15. Constants declared with `const` can be modified during runtime.
16. A `char` variable can store multiple characters at once.
17. `return 0;` typically indicates that the program finished successfully.
18. The modulus operator `%` can be used with `float` numbers.
19. `++x` (prefix) increments the value before using it in an expression.
20. Strings in C always end with the null character `\0`.

Part III — Correct the Code (5 Marks)

Identify and fix the error in each code snippet:

Snippet 1:

```
int main() {  
    int x = 10  
    printf("%d", x);  
    return 0;  
}
```

✓ Add a semicolon after 10 → `int x = 10;`

Snippet 2:

```
const int limit = 100;  
limit = 200;
```

✓ Cannot modify a **const variable**. Remove the assignment `limit = 200;`

Snippet 3:

```
if (age = 18) {  
    printf("Adult");  
}
```

✓ Use **==** instead of **=** → `if (age == 18)`

Snippet 4:

```
int score;  
scanf("%d", score);
```

✓ Add the address operator → `scanf("%d", &score);`

Snippet 5:

```
char grade = "A";
```

✓ Use single quotes for a character → `char grade = 'A';`

Part IV — Predict the Output (5 Marks)

What will be printed on the screen?

Code 1:

```
int x = 5;  
printf("%d", ++x + 2);
```

■ Output: 8

Code 2:

```
int i = 0;

while (i < 3) {

    printf("Hi ");

    i++;

}
```

■ Output: Hi Hi Hi

Code 3:

```
int a = 10, b = 5;

printf("%d", a > b && b < 2);
```

■ Output: 0

Code 4:

```
for (int i=1; i<=5; i++) {

    if (i == 3) break;

    printf("%d", i);

}
```

■ Output: 12

Code 5:

```
int x = 10;

x %= 3;

printf("%d", x);
```

■ Output: 1

■ Complete Answer Key

Check only after completing all parts!

Part I – MCQ Answers

Q1. C	Q2. B	Q3. C	Q4. C	Q5. D
Q6. B	Q7. C	Q8. C	Q9. C	Q10. C
Q11. B	Q12. B	Q13. B	Q14. C	Q15. B
Q16. C	Q17. C	Q18. B	Q19. B	Q20. C
Q21. B	Q22. C	Q23. B	Q24. C	Q25. B
Q26. C	Q27. C	Q28. B	Q29. B	Q30. C
Q31. C	Q32. B	Q33. A	Q34. C	Q35. B
Q36. B	Q37. C	Q38. B	Q39. B	Q40. C
Q41. D	Q42. B	Q43. C	Q44. B	Q45. B
Q46. A	Q47. B	Q48. C	Q49. B	Q50. A

Part II – True / False Answers

1. T	2. F	3. T	4. F	5. F
6. T	7. F	8. T	9. F	10. T
11. T	12. T	13. F	14. F	15. F
16. F	17. T	18. F	19. T	20. T

Part III – Correct the Code & Part IV – Predict the Output

III – 1	Add semicolon: <code>int x = 10;</code>
III – 2	Remove assignment – <code>const</code> cannot be modified.
III – 3	Use <code>==</code> : <code>if (age == 18)</code>
III – 4	Add <code>&</code> : <code>scanf("%d", &score);</code>
III – 5	Use single quotes: <code>char grade = 'A';</code>
IV – 1	Output: 8
IV – 2	Output: Hi Hi Hi
IV – 3	Output: 0
IV – 4	Output: 12
IV – 5	Output: 1

Good luck! You will ace the midterm! ■